

ex 17

English



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Run

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Output

Input

```
1 clear;
2 close all;
3 clc;
4
5 %% Parameters
6 N = 16;
7 d = 0.5;
8
9 th = -90:0.2:90;
10 th_r = th * pi / 180;
11
12 Ns = 3;
13 ang = [-20 0 30];
14
15 %% Steering Vector
16 a = @(t) exp(1j * 2 * pi * d * (0:N-1)' .* sin(t));
17
18 %% Steering Matrix
19 A = zeros(N,Ns);
20
21 for k = 1:Ns
22     A(:,k) = a(ang(k) * pi / 180);
23 end
24
25 %% Digital Beamforming
26 wdig = conj(A);
27
28 for k = 1:Ns
29     wdig(:,k) = wdig(:,k) / norm(wdig(:,k));
30 end
31
32 wdig = wdig / norm(wdig,'fro');
33
34 %% Analog Beamforming
35 wana = zeros(N,Ns);
36
37 for k = 1:Ns
```

STEERING PEAKS

Beam 1 : Peak angle = 20 degrees
Beam 2 : Peak angle = 0 degrees
Beam 3 : Peak angle = -30 degrees

Simulation completed successfully.

[Execution complete with exit code 0]

